PRAKTIKUM BASIS DATA

MODUL 9

“Scripting Procedure dan Function”

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JURNAL

Soal

Running code sql berikut, screenshot hasilnya dan analisis hasil code tersebut

1. create table employees as select * from hremployees.
2. select * from employees.
3.

```sql
CREATE OR REPLACE PROCEDURE increase_salary (
department_id_in   IN employees.department_id%TYPE,
    increase_pct_in    IN NUMBER)
     IS
BEGIN
    FOR employee_rec
            IN (SELECT employee_id
            FROM employees
           WHERE department_id = increase_salary.department_id_in)
    LOOP
            UPDATE employees emp
                SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
            WHERE emp.employee_id = employee_rec.employee_id;
            DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
```

4. BEGIN
   increase_salary (90, .50);
   ROLLBACK;
   END;
   /
   select * from employees where department_id = 90
5. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4.

Jawaban :

1.

```sql
--
CREATE TABLE hremployees (
    employee_id    NUMBER(6)       NOT NULL,
    first_name     VARCHAR2(20),
    last_name      VARCHAR2(25)  NOT NULL,
    email          VARCHAR2(25)  NOT NULL,
    phone_number   VARCHAR2(20),
    hire_date      DATE          NOT NULL,
    job_id         VARCHAR2(10)  NOT NULL,
    salary         NUMBER(8,2),
    commission_pct NUMBER(2,2),
    manager_id     NUMBER(6),
    department_id  NUMBER(4),
    CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)
);
```

Script Output × | Query Result ×

Task completed in 0.027 seconds

```
Table HREMPLOYEES created.
```

Worksheet | Query Builder

```sql
-- INSERT tambahan department lain (agar SELECT * tidak kosong)
INSERT INTO hremployees VALUES (103, 'Alexander', 'Hunold',  'AHUNOLD',  '590.423.4567', TO_DATE('03-JAN-2006','DD-MON-YYYY'), 'IT_PROG',  9000,  NULL
INSERT INTO hremployees VALUES (104, 'Bruce',     'Ernst',   'BERNST',   '590.423.4568', TO_DATE('21-MAY-2007','DD-MON-YYYY'), 'IT_PROG',  6000,  NULL
INSERT INTO hremployees VALUES (107, 'Diana',     'Lorentz', 'DLORENTZ', '590.423.5567', TO_DATE('07-FEB-2007','DD-MON-YYYY'), 'IT_PROG',  4200,  NULL
INSERT INTO hremployees VALUES (124, 'Kevin',     'Mourgos', 'KMOURGOS', '650.123.5234', TO_DATE('16-NOV-2007','DD-MON-YYYY'), 'ST_MAN',   5800,  NULL
INSERT INTO hremployees VALUES (141, 'Trenna',    'Rajs',    'TRAJS',    '650.121.8009', TO_DATE('17-OCT-2003','DD-MON-YYYY'), 'ST_CLERK', 3500,  NULL
INSERT INTO hremployees VALUES (149, 'Eleni',     'Zlotkey', 'EZLOTKEY', '011.44.1344.429018', TO_DATE('29-JAN-2008','DD-MON-YYYY'), 'SA_MAN',   10500
INSERT INTO hremployees VALUES (174, 'Ellen',     'Abel',    'EABEL',    '011.44.1644.429267', TO_DATE('11-MAY-2004','DD-MON-YYYY'), 'SA_REP',   11000
--
create table employees as select * from hremployees;

select * from hremployees;
--
SET SERVEROUTPUT ON;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 10 in 0.005 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-01-2006	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-05-2007	IT_PROG	6000	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-02-2007	IT_PROG	4200	(null)	103	60
7	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-11-2007	ST_MAN	5800	(null)	100	50
8	141	Trenna	Rajs	TRAJS	650.121.8009	17-10-2003	ST_CLERK	3500	(null)	124	50
9	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-01-2008	SA_MAN	10500	0,2	100	80
10	174	Ellen	Abel	EABEL	011.44.1644.429267	11-05-2004	SA_REP	11000	0,3	149	80

2.

```
select * from employees;
--
SET SERVEROUTPUT ON;
```

Script Output | Query Result

SQL | All Rows Fetched: 12 in 0.003 seconds

	EMPLOYEE_ID	LAST_NAME	JOB_ID	SALARY	DEPARTMENT_ID
1	141	Nayer	ST_CLERK	3200	50
2	142	Mikkilineni	ST_CLERK	2700	50
3	143	Olson	ST_CLERK	2600	50
4	144	Bissot	ST_CLERK	3300	50
5	145	Atkinson	ST_CLERK	2800	50
6	146	Mallin	ST_CLERK	3300	60
7	147	Rogers	ST_CLERK	2900	60
8	148	Ladwig	ST_CLERK	3600	60
9	149	Stiles	ST_CLERK	3200	80
10	150	Seo	ST_CLERK	2700	80
11	151	Rajs	ST_CLERK	3500	80
12	152	Davies	ST_CLERK	3100	80

3 dan 4

```
--
CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in IN employees.department_id%TYPE,
    increase_pct_in  IN NUMBER
)
IS
BEGIN
    UPDATE employees
    SET salary = salary + (salary * increase_pct_in / 100)
    WHERE department_id = department_id_in;

    DBMS_OUTPUT.PUT_LINE('Updated rows: ' || SQL%ROWCOUNT);
END;
/
--
```

Script Output | Query Result

Task completed in 0.027 seconds

```
or drop the existing object so it can be reused.

Procedure INCREASE_SALARY compiled

Updated rows: 0


PL/SQL procedure successfully completed.
```

5. jika rollback dihapus :

- Perubahan gaji tetap tersimpan (commit otomatis saat session selesai / commit manual)
- Salary di department 90 akan benar-benar naik 50%